

ROLL NO:250701573

EXPERIMENT NO:16 TIME SERIES ANALYSIS OF WEBSITE TRAFFIC

PROGRAM:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose
data = pd.read_csv('website_traffic_modified.csv',
                  parse_dates=['Date'],
                  index_col='Date')

data.head()

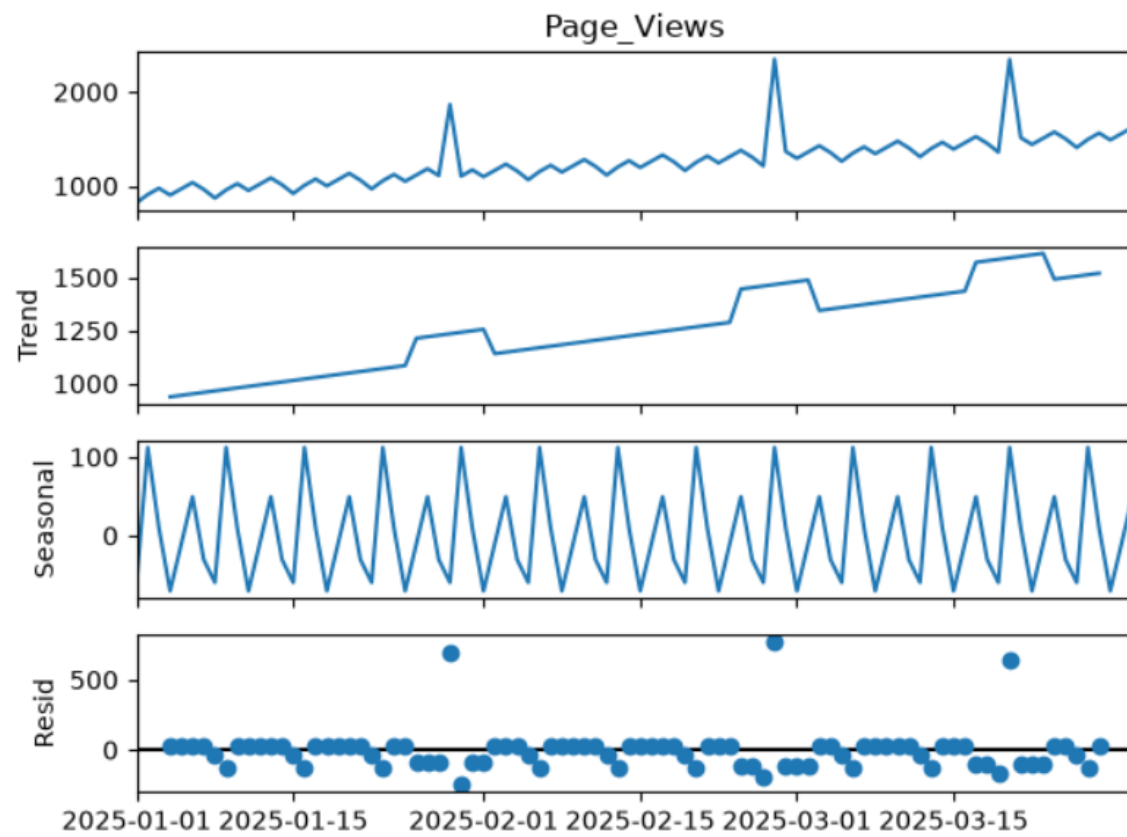
decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()

plt.show()
```

OUTPUT:

Page_Views	Unique_Visitors	Bounce_Rate	
Date			
2025-01-01	820	510	60.00
2025-01-02	907	549	57.92
2025-01-03	974	578	56.84
2025-01-04	901	547	58.76
2025-01-05	968	576	55.68



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